

Sales Analysis Dashboard using Power BI

Sales Analysis Dashboard using Power BI

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Subject: Data Analytics / Business Intelligence

Tool Used: Microsoft Power BI

Dataset: Sales Analysis Dataset

Introduction

Business organizations generate large amounts of sales data every day. Analyzing this data manually is difficult and time-consuming. Power BI provides interactive visualization and analytical tools that help convert raw data into meaningful business insights.

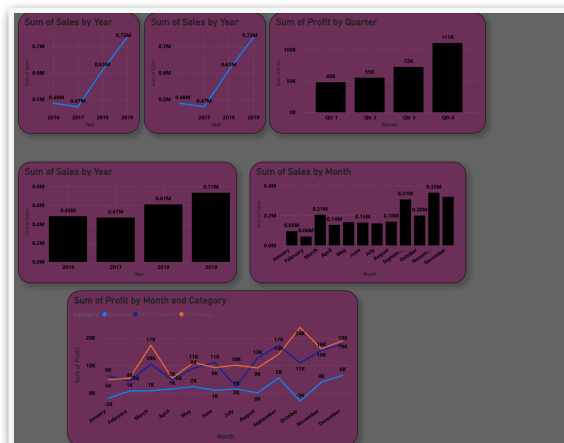
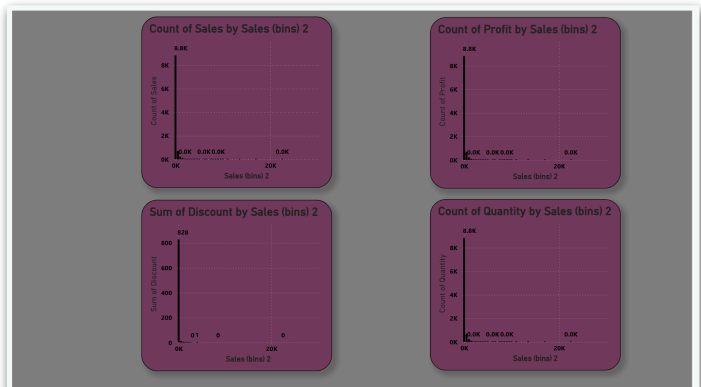
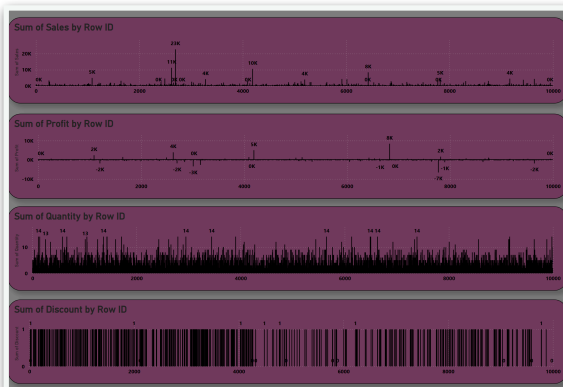
This project focuses on analyzing sales data using Power BI to identify trends, category performance, customer behavior, profitability, forecasting patterns, and business anomalies. Different visualizations and analytical techniques were used to understand the dataset and support business decision-making.

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Item ID	Order ID	Buy Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit						
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Objectives

The major objectives of this project are:

- To analyze sales and profit trends.
- To identify top-performing and low-performing categories.
- To examine categorical and numerical distributions.
- To study relationships between sales, profit, and discounts.
- To detect outliers and anomalies in the dataset.
- To forecast future sales trends using historical data.
- To create an interactive dashboard for business insights.



Requirement Analysis

Software Requirements

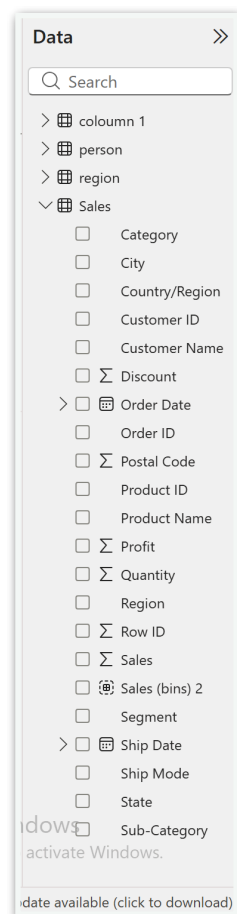
- Microsoft Power BI Desktop
- Microsoft Excel
- Windows Operating System

Hardware Requirements

- Minimum 4 GB RAM
- Intel i3 Processor or above
- 5 GB free storage

Dataset Fields Used

- Sales
- Profit
- Quantity
- Discount
- Category
- Sub-Category
- Region
- Segment
- Order Date



Methodology

The project followed a structured analytical workflow:

1. Data Collection
2. Data Cleaning
3. Data Transformation
4. Data Visualization
5. Trend Analysis
6. Outlier Detection
7. Forecasting
8. Insight Generation

Power BI visuals such as bar charts, scatter plots, line charts, matrix visuals, decomposition trees, and forecasting charts were used for analysis.

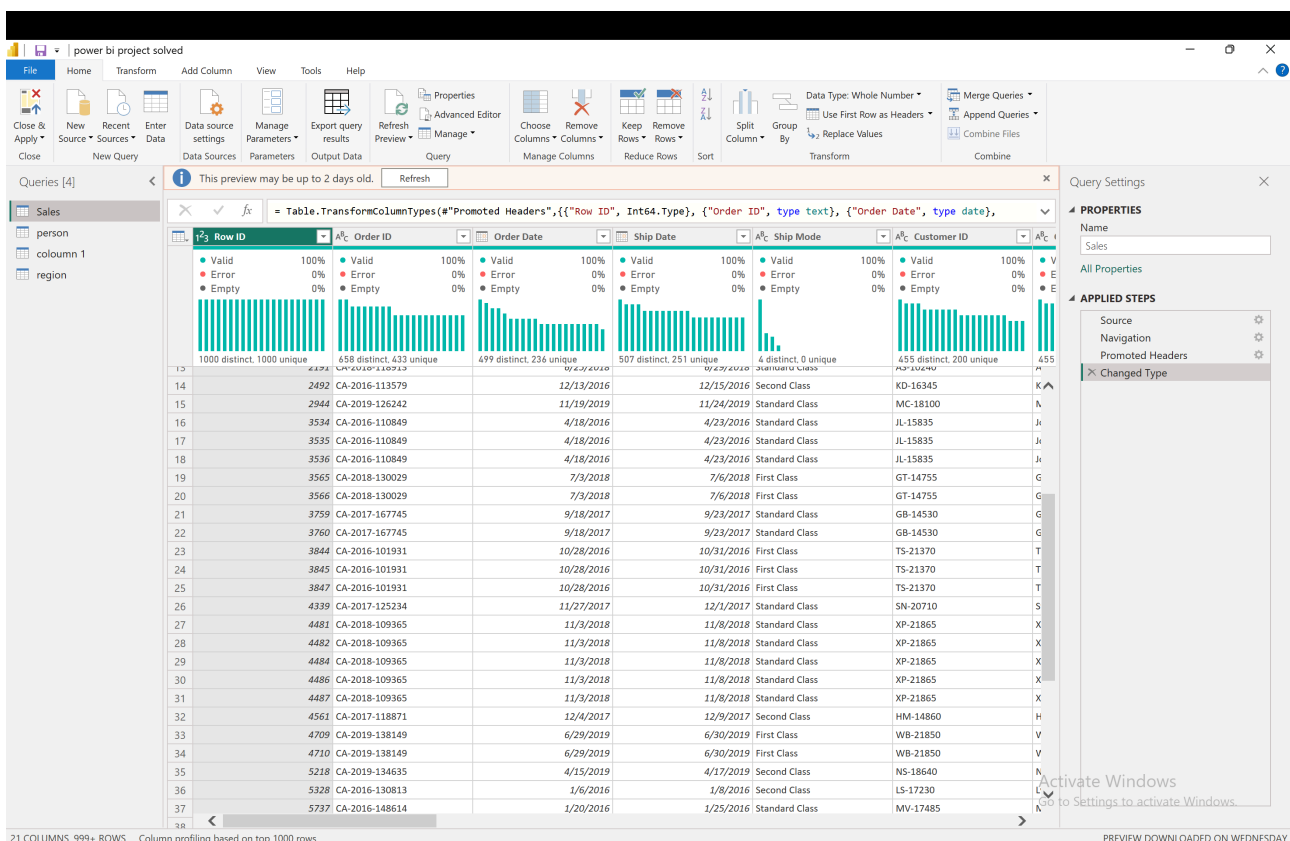
Data Cleaning and Transformation

Before analysis, the dataset was cleaned and transformed using Power Query Editor.

The following preprocessing operations were performed:

- Removal of duplicate values
- Identification of missing values
- Verification of unique and distinct values
- Creation of bins for histogram analysis
- Formatting date columns
- Data aggregation and filtering

These transformations improved data quality and ensured accurate analysis.



Average Value Analysis

Average analysis was performed on numerical columns such as Sales, Profit, Quantity, and Discount.

Card visuals and table visualizations were used to display average values.

This analysis helped understand the general magnitude and central tendency of the dataset.

Insights

- Average sales values indicated overall transaction size.
- Profit averages helped evaluate business profitability.
- Discount averages revealed pricing and promotional patterns.

Average of Sales	Average of Quantity	Average of Profit	Average of Discount
229.86	3.79	28.66	0.09

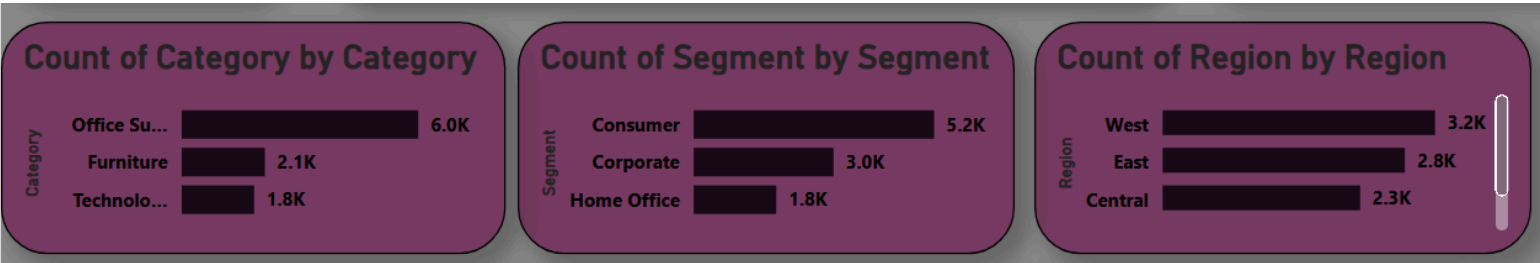
Distribution of Categorical Values

Bar charts were created to analyze the distribution of categorical values such as Category, Region, Segment, and Ship Mode.

Count aggregation was used to identify the frequency of records in each category.

Insights

- Certain categories dominated the dataset.
- Some regions contributed more transactions than others.
- Customer segments showed varying purchasing patterns.



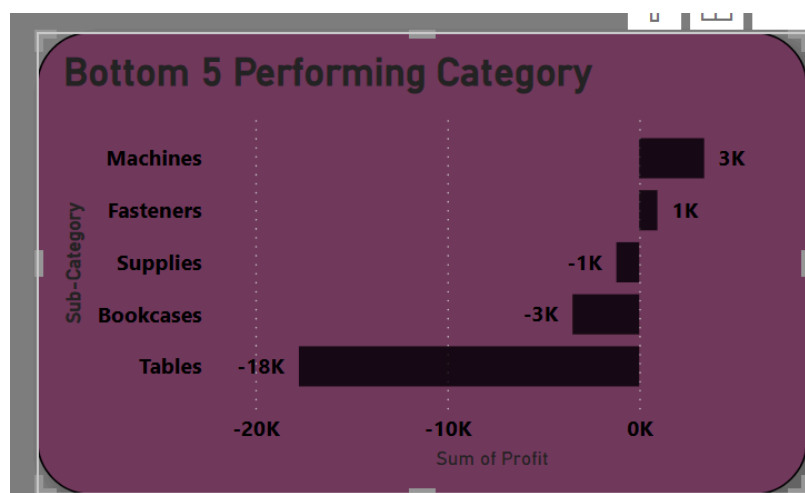
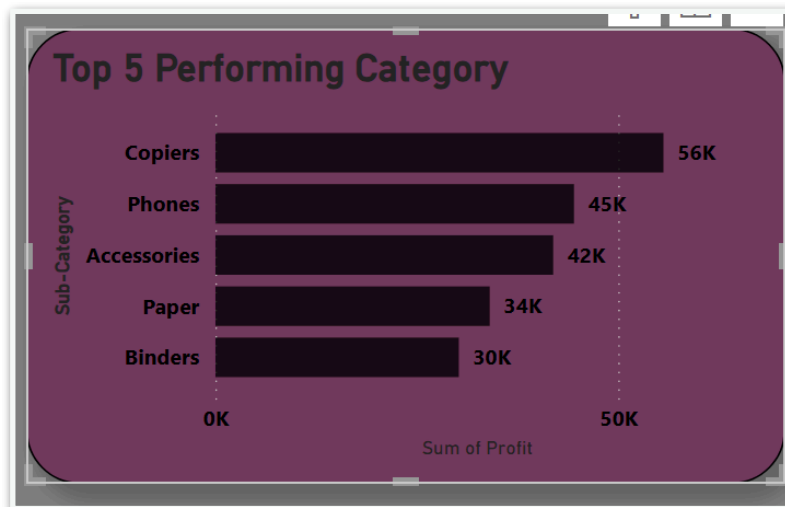
Top and Bottom Performing Categories

Top N filtering was used to identify the highest and lowest performing categories based on Profit and Sales.

Clustered bar charts helped compare category performance.

Insights

- Technology-related categories generated high sales.
- Some sub-categories showed consistently low profitability.
- Low-performing categories may require business improvement strategies.



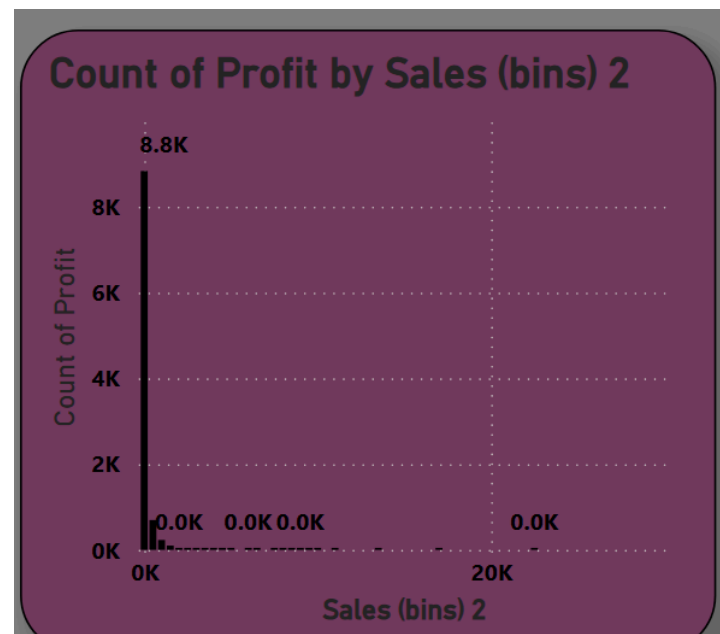
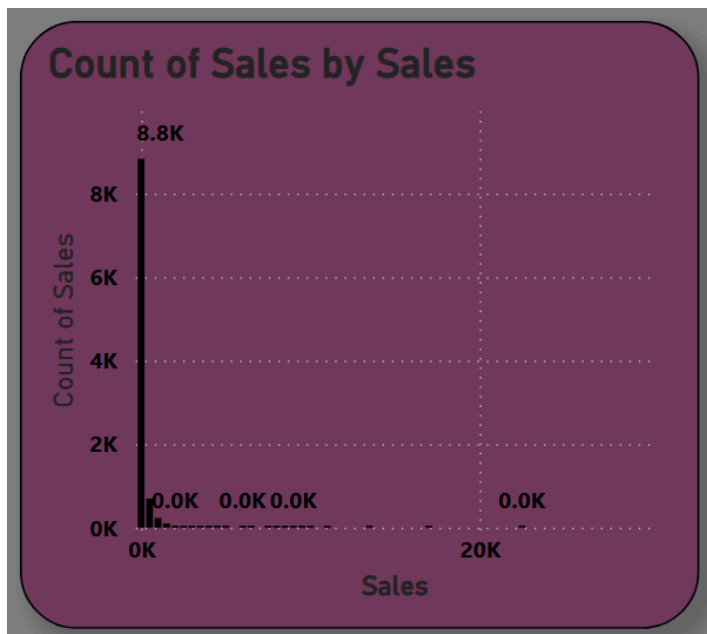
Histogram and Range Analysis

Histogram analysis was performed using bins to understand how numerical values were distributed across intervals.

Clustered column charts were used to display frequencies of records within numerical ranges.

Insights

- Most sales values were concentrated within lower ranges.
- Only a few records had extremely high values.
- Data distributions were positively skewed.



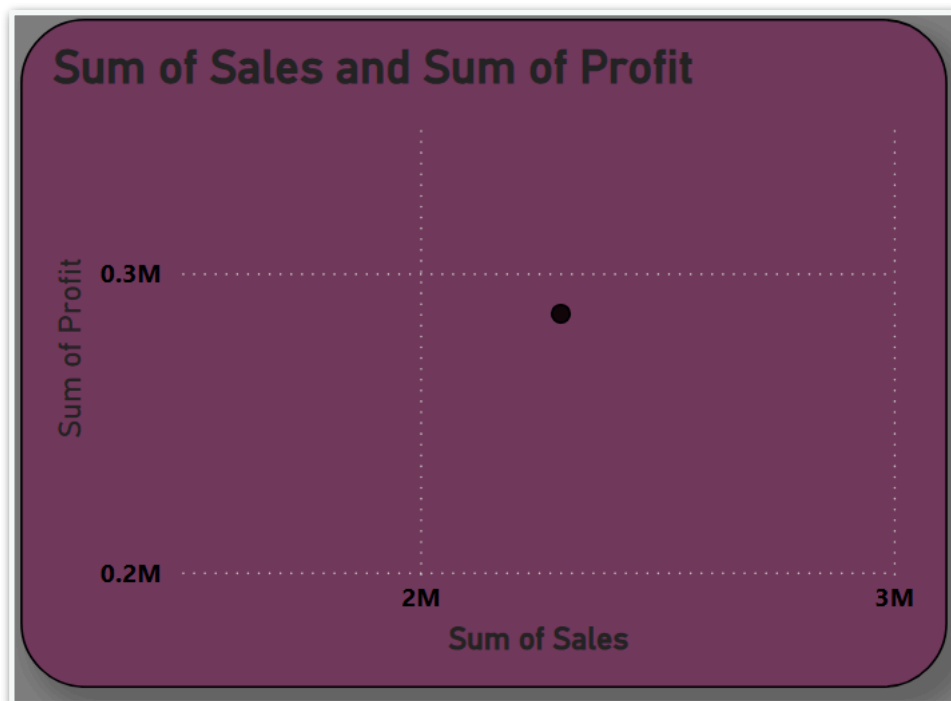
Correlation Analysis using Scatter Plot

Scatter plots were used to analyze the relationship between Sales and Profit.

Each point in the scatter plot represented an individual transaction.

Insights

- A positive correlation was observed between Sales and Profit.
- Several transactions generated high sales with negative profit.
- Outlier points indicated unusual business behavior.



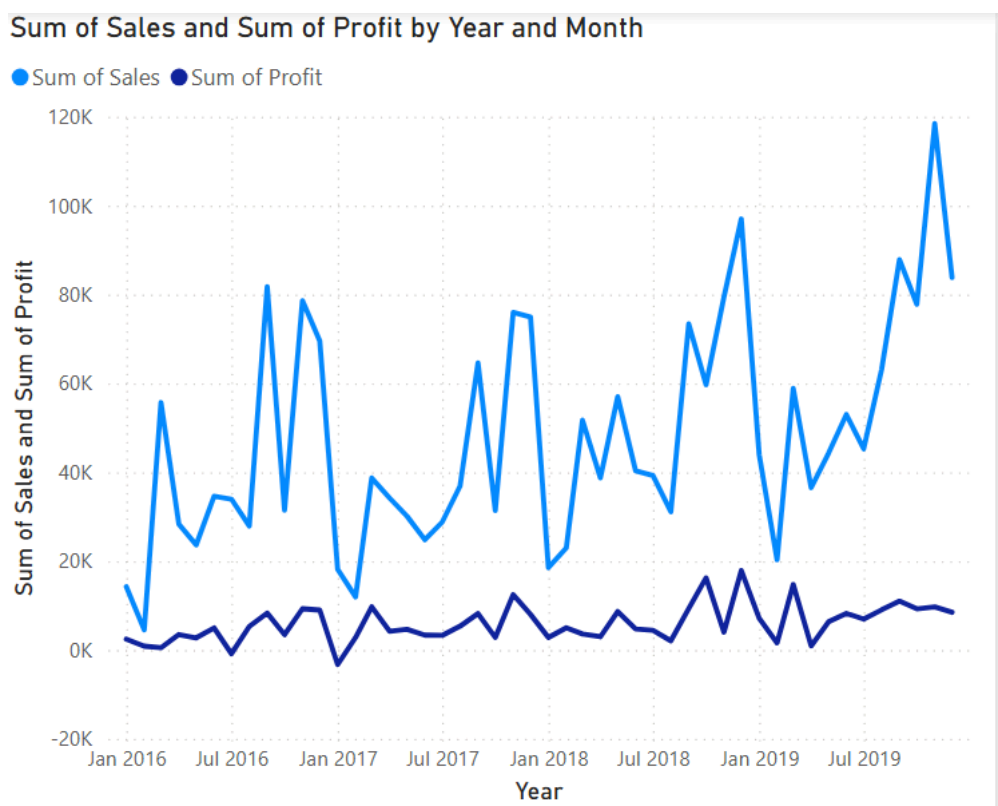
Trend Analysis Over Time

Line charts were used to analyze sales and profit trends over time using the Order Date column.

Monthly and yearly trends were observed to identify business growth patterns.

Insights

- Sales showed fluctuations across different periods.
- Certain months demonstrated higher performance.
- Trend analysis helped understand business growth over time.



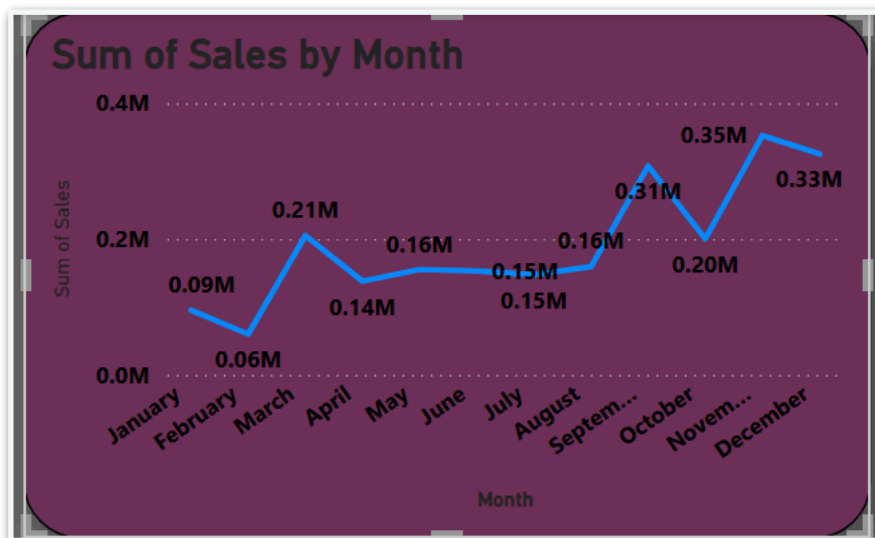
Seasonal and Periodic Pattern Analysis

Seasonality analysis was performed to identify repeating patterns in sales performance across different months.

Line charts helped identify recurring increases and decreases in sales.

Insights

- Certain periods showed recurring sales peaks.
- Seasonal fluctuations affected business performance.
- Time-based analysis helped identify recurring customer behavior.



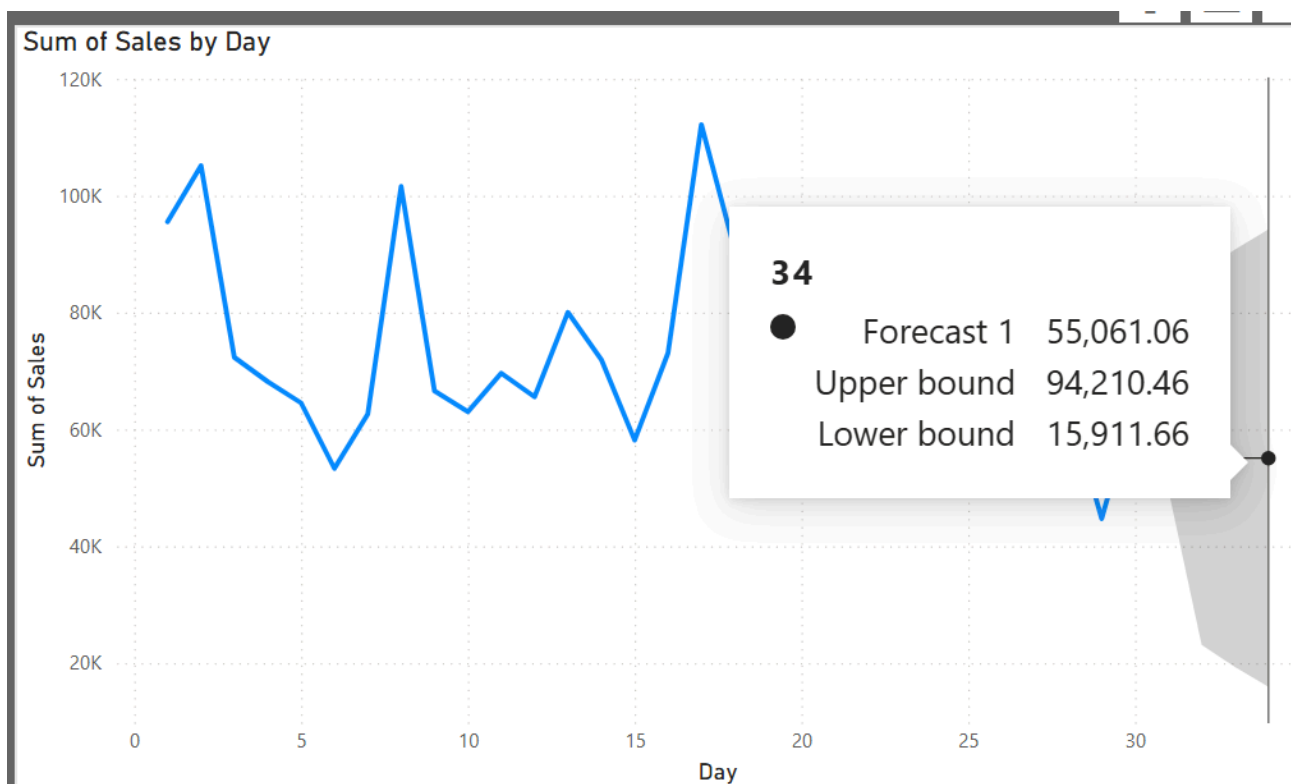
Forecasting Analysis

Power BI forecasting functionality was used to predict future sales trends based on historical data.

Forecast intervals, upper bounds, and lower bounds were analyzed.

Insights

- Sales are expected to continue following historical trends.
- Forecast ranges provide estimated future uncertainty.
- Forecasting supports future planning and decision-making.



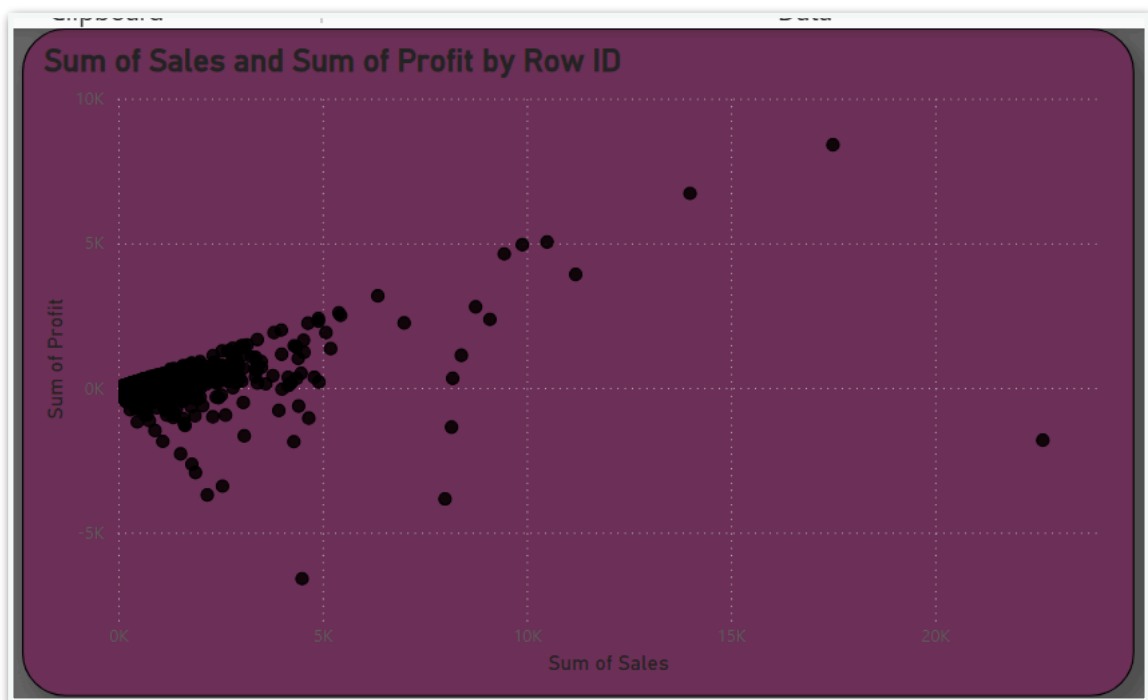
Outlier Detection Analysis

Scatter plots were used to identify outliers and anomalies in Sales and Profit values.

Outliers represent unusual transactions that significantly deviate from normal patterns.

Insights

- Some orders generated very high sales but negative profit.
- Negative-profit outliers may indicate excessive discounts.
- Outlier analysis helps detect risks and opportunities.



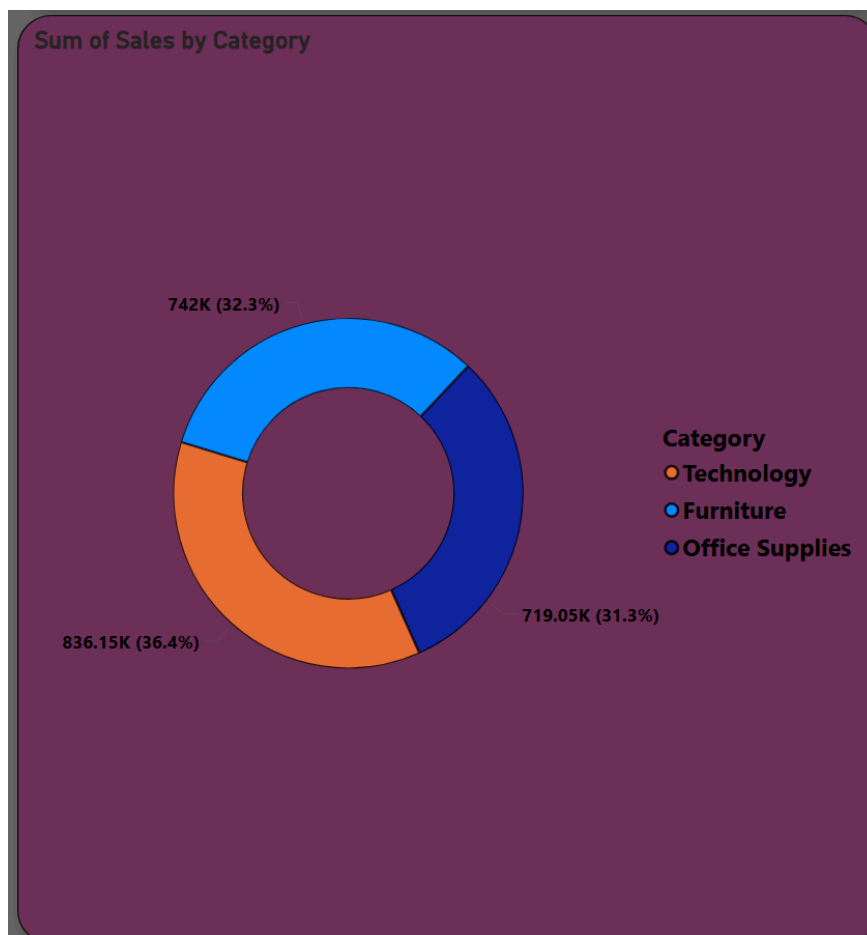
Contribution Analysis

Donut charts were used to identify the percentage contribution of each category to total sales.

This analysis helped understand the relative importance of each category.

Insights

- Technology category contributed the highest share.
- Some categories had smaller contributions to total revenue.
- Contribution analysis helps prioritize business focus areas.



Hierarchical Analysis using Matrix Visual

Matrix visuals were used to analyze hierarchical relationships such as Category → Sub-Category.

Drill-down functionality enabled deeper exploration of business performance.

Insights

- Certain sub-categories dominated overall sales.
- Some regions and states contributed significantly more revenue.
- Hierarchical analysis helped identify detailed contributors.

Category	Sum of Sales
Furniture	741,999.80
Bookcases	114,880.00
Chairs	328,449.10
Furnishings	91,705.16
Tables	206,965.53
Office Supplies	719,047.03
Appliances	107,532.16
Art	27,118.79
Binders	203,412.73
Envelopes	16,476.40
Fasteners	3,024.28
Labels	12,486.31
Paper	78,479.21
Storage	223,843.61
Supplies	46,673.54
Technology	836,154.03
Accessories	167,380.32
Copiers	149,528.03
Machines	189,238.63
Phones	330,007.05
Total	2,297,200.86

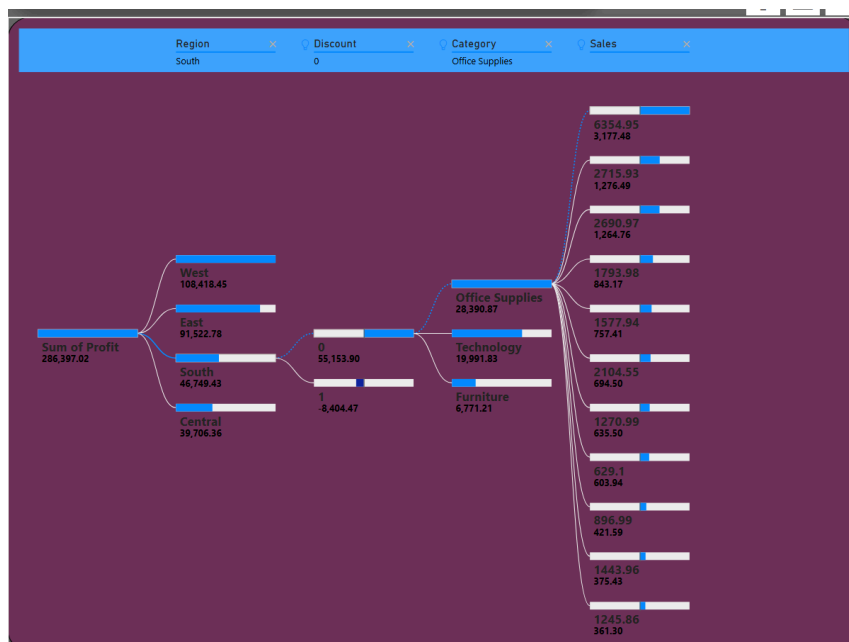
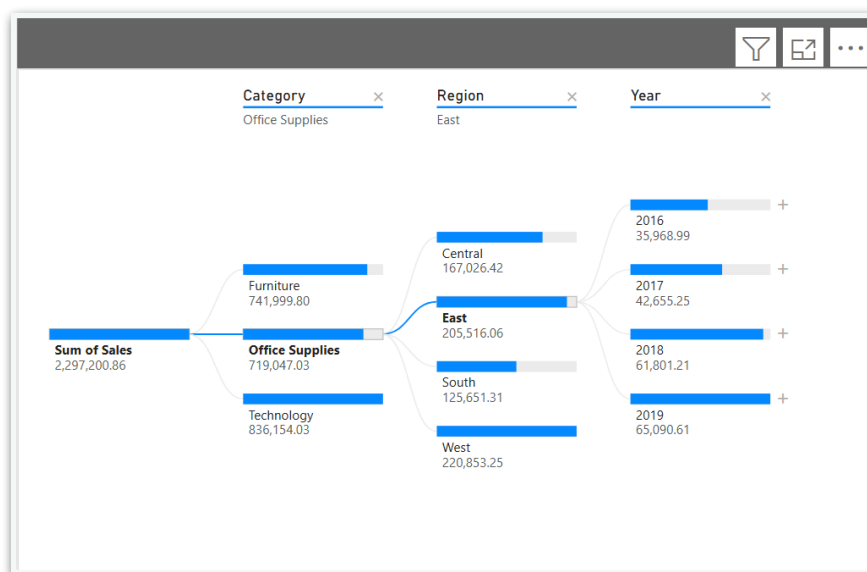
Decomposition Tree Analysis

The decomposition tree visual was used to identify how different variables interactively affect Profit and Sales.

Variables such as Region, Category, State, and Discount were used for hierarchical breakdown analysis.

Insights

- Certain regions generated higher profitability.
- Discount levels strongly influenced profit margins.
- Multi-level relationships affected business outcomes.



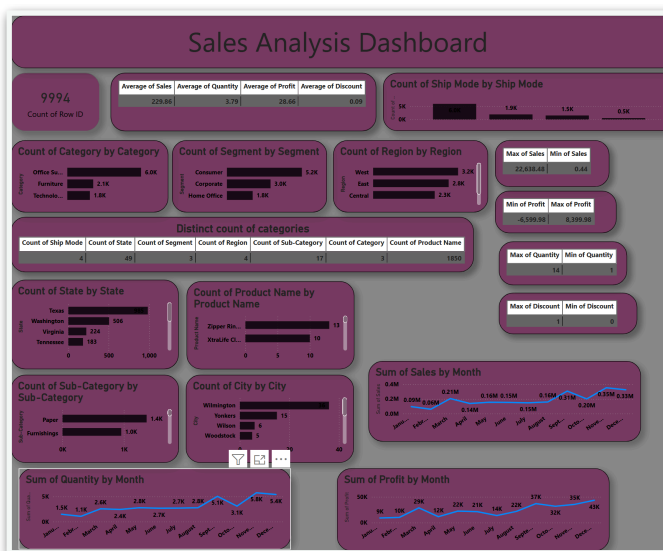
Key Findings and Recommendations

Key Findings

- Technology category generated the highest sales.
- Several high-sales orders resulted in negative profit.
- Sales trends increased during specific seasonal periods.
- Discounts negatively affected profitability in multiple cases.
- Forecasting indicated continued future sales growth.

Recommendations

- Reduce excessive discounting.
- Focus on high-performing categories.
- Monitor loss-making transactions.
- Improve low-performing regions.
- Use forecasting for future inventory and sales planning.



Conclusion

The Sales Analysis Dashboard successfully demonstrated how Power BI can transform raw sales data into meaningful business insights.

The project included data cleaning, transformation, visualization, forecasting, outlier detection, hierarchical analysis, and trend analysis.

Interactive dashboards helped identify important business patterns, high-performing categories, seasonal trends, and profitability issues.

Power BI proved to be an effective tool for business intelligence, reporting, and data-driven decision-making.